

Process-Oriented Evaluation of Large Language Models in RAG Systems Powered by Association Rules for Environmental Protection Data

Grit Behrens¹ [0009-0009-0247-8204], Kostas Karatzas² [0000-0002-1033-5985], Cezary Orłowski³ [0000-0001-7716-5522], Leon Unruh, Rafał Grzeszczyk³ [0000-0003-0421-9184]

¹HSBI Bielefeld University of Applied Sciences and Art, grit.behrens@hsbi.de

²Aristotle University of Thessaloniki, kkara@auth.gr

³WSB Merito University in Gdansk, cezary.orlowski@gdansk.merito.pl,
rafal.grzeszczyk@gdansk.merito.pl

Abstract. The rapid development of large language models (LLMs) creates new opportunities for the analysis and interpretation of environmental data, while simultaneously introducing significant challenges related to the evaluation of generated responses. This paper proposes a process-oriented evaluation model for language models within a Retrieval-Augmented Generation (RAG) system, where domain knowledge is represented in the form of association rules and stored in a vector database. Three different language models were comparatively evaluated using an identical knowledge base and the same set of queries. A multimodal model (LMM) was employed as an evaluation mechanism, acting as a meta-evaluator under the LLM-as-a-Judge paradigm. The experimental results enabled the unambiguous identification of the model best suited for further analysis of environmental protection data. The proposed approach constitutes a formal and reproducible method for selecting language models in RAG systems with high domain-specific requirements.

Keywords: Retrieval-Augmented Generation, language models, LLM-as-a-Judge, association rules, environmental protection data, explainable AI

1 Introduction

The application of large language models (LLMs), most commonly transformer-based, to environmental data and information analysis, is gaining importance with the growing availability of unstructured knowledge sources such as environmental reports, environmental quality indicators, and relevant regulatory documents [1]. LLMs can combine semantic retrieval with structured forms of reasoning to support tasks that resemble “interpretation” of complex material, such as synthesising evidence, comparing statements across sources, and producing concise analytic narratives. At the same time, a critical challenge emerges: selecting a model that not only produces linguistically fluent responses but also maintains high domain-specific factual accuracy—an issue that is

particularly acute in environmental contexts where outputs may inform compliance-related or risk-sensitive decisions.

A technical response to the accuracy limitations of standalone LLMs is retrieval-augmented generation (RAG), which integrates a retriever over an external knowledge store with a generative model that conditions on retrieved evidence [2]. RAG was introduced to reduce reliance on internalised parametric memory and to improve factuality by grounding outputs in accessible sources. Subsequent work in open-domain question answering demonstrated that performance can improve when models are provided with more—and more relevant—retrieved passages, underscoring the practical importance of retrieval quality and evidence aggregation [3]. These findings are directly relevant to environmental protection settings, where information (a) can be related to multiple environmental domains of interest (air, water quality, noise, waste, etc.), (b) is included in heterogeneous (in terms of terminology, metadata structure, format, completeness and regulatory relevance) sources (c) is fragmented across diverse document types, while retrieval must cope with challenges related to measurement contexts, and jurisdictional specificity.

Most RAG implementations remain document- or passage-centric: they index text segments and treat retrieval as the selection of supporting excerpts. The present study explores an alternative design choice that is motivated by the needs of environmental analysis and explainability. Instead of using documents as the primary unit of knowledge representation, it uses association rules that encode explicit relationships among variables, conditions, and outcomes. This shift aims to make the “evidence” used by the generator more structured and directly inspectable than free-form text alone. In practical terms, a rule-based knowledge structure supports retrieval that is better aligned with causal or correlational queries (e.g., links between environmental pressures and observed indicators), while also identifying artefacts that are easier to audit within a formal evaluation procedure.

The move from standalone generation to retrieval-grounded pipelines changes what evaluation must capture. Performance can no longer be judged solely by output fluency or “agreement” with reference texts. A credible evaluation must account for (i) whether the retriever returns evidence that is relevant and sufficiently complete, (ii) whether the generator’s output remains faithful to retrieved items, and (iii) whether the end-to-end system behaves consistently across query types and levels of evidential ambiguity. This is particularly important for environmental protection data, where uncertainty and heterogeneity are common and where an apparently plausible response can still be misleading if it overgeneralises or misattributes evidence.

Alongside the development of RAG, evaluation has progressed beyond classical text-overlap measures (in automatic evaluation setting), which often fail to distinguish well-phrased errors from correct, grounded answers, and beyond human evaluation, which acts as a bottleneck when massive evaluation is necessary. One emerging direction is LLM-as-a-Judge, in which a language model is used as an evaluator of another model’s output [4][5]. Environmental decision-support operates under regulatory conditions and societal expectations that often require explanations that are intelligible to non-specialists and defensible in audit contexts. The broader XAI literature argues that

interpretability should be treated as a scientific and design requirement rather than a cosmetic add-on [6].

In the literature on the evaluation of systems based on large language models (LLMs) in RAG architectures, four complementary, yet still partially separate, research streams can be clearly distinguished. In the area of LLM-based generation evaluation, works [7][8][9] propose a set of metrics such as faithfulness, answer relevance, and context precision, which allow for automatic assessment of the degree of grounding of LLM responses in the provided context; however, this method remains limited to the model–context relationship and does not take into account external knowledge structures, such as domain rules. In turn, in the integrative evaluation stream based on the neuro-symbolic approach (neuro-symbolic integration and constraint-based evaluation), represented, among others, by [10][11][12] analyzes combining neural models with symbolic knowledge by imposing logical constraints or rules, which allows for verification of the consistency of responses with formal knowledge, but typically does not provide precise, quantitative metrics of generation quality characteristic for RAG systems. In the context of end-to-end evaluation of pipelines for retrieval-augmented generation systems (end-to-end RAG pipeline evaluation), publications [13][14][15] systematize metrics such as Exact Match, F1, or latency, indicating the need for a comprehensive evaluation of RAG systems. However, they focus mainly on the efficiency and accuracy of responses, omitting the aspect of compliance with domain knowledge and mining rules. In contrast, the uncertainty-aware and calibration-based evaluation of LLMs, as in [16][17][18], introduces mechanisms for estimating uncertainty and calibrating LLM responses, enabling the assessment of prediction reliability, but does not integrate this information with semantic consistency with rules or expert knowledge. Analysis of these approaches reveals a significant research gap: the lack of a unified evaluation mechanism that simultaneously measures the grounding of responses in context (faithfulness) and verifies their consistency with association rules or symbolic knowledge (rule consistency). In particular, generational evaluation methods provide precise response quality metrics but ignore global dependencies in the data, while neuro-symbolic approaches ensure logical consistency but do not offer granular assessment of response quality in the context of RAG.

Consequently, it is justified to propose a hybrid evaluation model that builds on the LLM-as-a-judge paradigm while incorporating the neuro-symbolic approach by using association rules as formal evaluation constraints. In this approach, the language model acts as a meta-evaluator, not only assessing the response for contextual consistency (faithfulness, relevance) but also interpreting and verifying its compliance with association rules extracted from environmental data (e.g., relationships between pollution levels), assigning semantic meaning to them, and detecting violations. This hybrid, therefore, combines the LLM's ability to assess generation quality with the formal validation of hidden knowledge in the data, enabling the simultaneous detection of contextual hallucinations and inconsistencies with empirical patterns, representing an important direction for the development of more reliable and domain-specific RAG systems.

2 Materials and Methods

The construction of a process-oriented evaluation model for language models operating on environmental protection data is conditioned by several methodological and domain-specific constraints. Environmental knowledge is inherently heterogeneous, context-dependent, and often embedded in regulatory documents or expert-driven narratives, that limit the applicability of purely data-driven or surface-level evaluation approaches. Consequently, the proposed methodology assumes that both knowledge representation and model assessment must explicitly account for domain semantics and interpretability.

A second conditioning factor is the architectural complexity of Retrieval-Augmented Generation systems, in which multiple components—knowledge extraction, vector retrieval, and generative inference—jointly influence the final output. In such systems, evaluating a language model in isolation is insufficient, as model performance is tightly coupled with the structure and quality of the retrieved knowledge. This motivated the adoption of a process-based evaluation perspective, in which all models are assessed under identical retrieval conditions derived from a shared, rule-based knowledge representation.

Finally, the lack of reliable, human-scalable evaluation mechanisms for domain-specific generative tasks necessitates the use of automated yet semantically informed assessment strategies. The methodology therefore incorporates a multimodal language model acting as a meta-evaluator within the LLM-as-a-Judge paradigm. This design choice enables consistent, repeatable, and context-aware comparison of multiple language models, forming the foundation for the process model detailed in the following sections.

2.1 Construction of the Process Model

The process model proposed in this study was designed to ensure a controlled, reproducible, and domain-aware evaluation of language models applied to environmental protection data. Its construction was guided by three fundamental assumptions: (i) domain knowledge must be explicitly structured and interpretable, (ii) all evaluated language models must operate under identical retrieval conditions, and (iii) model assessment must go beyond surface-level linguistic quality and address semantic and analytical correctness. These assumptions directly informed the selection of materials, the representation of knowledge, and the overall system architecture.

The research material consisted of a thematically coherent corpus of texts related to environmental protection, including descriptions of environmental processes, anthropogenic pressures, environmental quality indicators, and mitigation measures. The coherence of the corpus enabled the extraction of stable conceptual relationships rather than isolated factual statements. This characteristic was essential for constructing a knowledge base that captures recurring environmental dependencies and supports meaningful semantic retrieval.

The process model is designed to provide a structured and reproducible framework for evaluating language models within a Retrieval-Augmented Generation (RAG)

system, using environmental protection data as the target domain. The curated corpus of environmental texts is systematically analyzed to extract key concepts and relationships. These relationships are formalized as association rules ($A \rightarrow B$), where A and B denote co-occurring environmental concepts, capturing also dependencies among them.. These rules provided a reduced yet interpretable abstraction of domain knowledge, balancing semantic expressiveness with computational efficiency. The resulting rule set constituted the sole knowledge source used during retrieval, ensuring that all evaluated models relied on the same structured information.

The association rules were subsequently converted into vector embeddings and stored in a vector database, forming the retrieval backbone of the RAG system. The system architecture comprised three layers: (1) a query layer responsible for handling user questions $q \in Q$, where Q the set of all user questions, (2) a vector retrieval layer performing semantic search over the embedded association rules, and (3) a generative layer implemented by a language model. When a query is submitted, the system performs a semantic retrieval over the embedded rules, returning relevant knowledge to the generative layer. Three different language models, denoted as LLM_i, $i \in \{1,2,3\}$, were integrated separately into the generative layer, while sharing the same retrieval component to ensure experimental comparability. This shared retrieval ensures that all models are evaluated under identical informational conditions, isolating the effects of model architecture and generation strategy on output quality.

The evaluation component of the process model was implemented using a multimodal language model (LMM) acting as a meta-evaluator within the LLM-as-a-Judge paradigm. The LMM did not perform consensus aggregation but independently assessed the responses (answer_i, $i \in \{1,2,3\}$) generated by each language model. Evaluation was conducted according to four predefined criteria: C1 factual correctness, C2 completeness of the response, C3 logical coherence, and C4 analytical usefulness. The assessment was ordinal in nature and resulted in a ranked ordering of the evaluated models. Based on these evaluations, the LMM produces a ranked ordering of the models, identifying the most suitable candidate for deployment in environmental knowledge tasks. This structured evaluation process allows for consistent, repeatable comparison of models and supports the selection of a high-performing language model while maintaining transparency and traceability of the decision-making process. The whole procedure is visualized in fig 1., while the overall evaluation workflow is summarized below:

```

for each question q in Q:
  for each model LLMi:
    answeri = RAG(LLMi, q)
    for i=1:3
      for j=1:4
        (Cji)=LMM_Judge(answeri)
      scoresi = (C1i,C2i,C3i,C4i)
    end for;end for
  update_for each question q in Q:
    update_ranking(scores)
select best_LLMranking(scores)=rank(scores)

```

select best_LLM

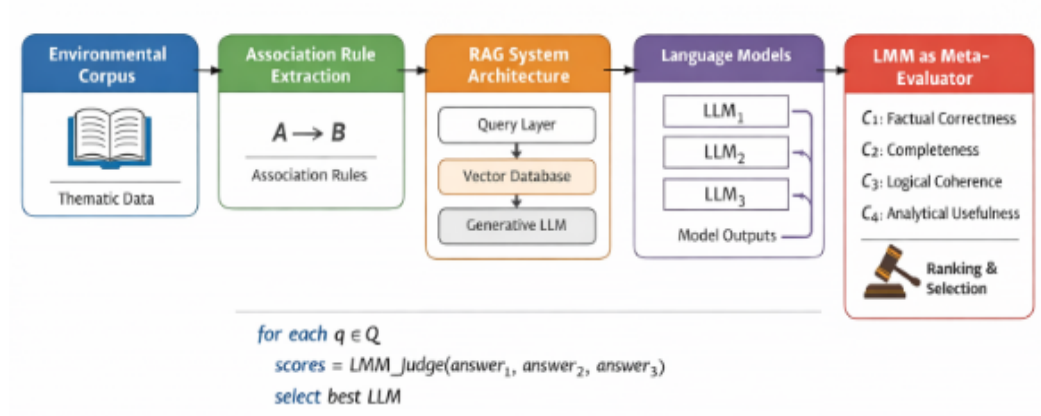


Fig. 1. Model of the data assessment

2.2 Model Assessment and Verification

The experimental process is designed to systematically evaluate multiple language models on environmental protection data using a structured, rule-based approach (fig. 2).

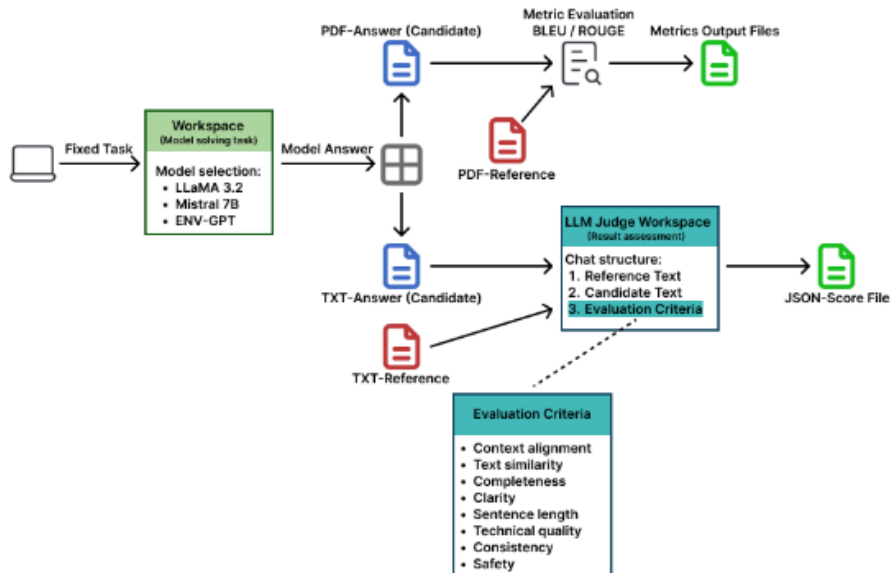


Fig. 2. Experimental Evaluation

Three language models—LLaMa 3.2 (for LLM1), Mistral 7B (for LLM2), and ENV_GPT (for LLM3)—independently generate candidate outputs based on the retrieved information. For each query, BLEU and ROUGE metrics are employed to produce initial candidate files, measuring surface-level similarity to a pre-defined reference file.

A meta-evaluator model (LMM) subsequently assesses these candidate outputs against the reference file using a set of domain-specific criteria: context alignment, text similarity, completeness, clarity, sentence length, technical quality, consistency, and safety. Each candidate file is scored according to these dimensions, allowing the system to rank the models in terms of overall performance. By integrating rule-based knowledge, multiple generative models, traditional text-similarity metrics, and a meta-evaluator, the process ensures a rigorous, reproducible, and transparent assessment of language models in environmental applications.

3 Results

The study was designed as a multi-stage process to evaluate the ability of LLMs to transform descriptive text into formal association rules. The starting point was a baseline text on air quality in urban environments, which includes a definition of the phenomenon, identification of key pollutants (NO_x, CO, VOC, PM, O₃, SO₂), their classification, and dependencies resulting from human activity. This text served as a reference source of knowledge against which the generated rules were assessed for accuracy and completeness.

Stage I - The reference text. In the first stage, the analysis was shortened and structured to highlight key cause-and-effect relationships. In particular, the study considered the relationships between emissions and pollutant sources, the impact of mobility patterns on NO_x and CO levels, and seasonal dependencies related to heating and meteorological conditions. For example, it was shown that SO₂ and particulate matter emissions increase in winter (therefore relevant concentrations are expected to be high), while the share of ozone within the gaseous pollutant concentrations increases in summer (as a result of photochemical reaction mechanisms).

Stage II - Prompt for creating association rules based on the reference text. In the second stage, a detailed association rule generation prompt was developed to enforce a clear structure for the response. The instructions included entries such as "Rule X: [Title]", "Antecedent : ...", "Consequent : ...", and "Rule statement : If ..., then ...". An example of the expected transformation involved a transition from a descriptive sentence to a rule, e.g. from a description of an issue to the form: "If fossil fuel combustion occurs, then sulfur dioxide levels increase".

Stage III - Association rules generated based on language models. In the third stage, the language models generated sets of association rules, analysing the reference text. On this basis, the llama3.2_3b model generated, for example, the following two rules:

- A. "If vehicle emissions and industrial processes occur" (this is the condition), "then air pollution is likely to be present" (this is the corresponding result) which correctly maps the relationship between cause and effect .
- B. "If urban air quality needs to be evaluated" (this is the condition), "then reference-grade monitoring or low-cost sensing can be used" (this is the corresponding method choice), which is the consequent part associated with the fragment concerning methods grades .

Additionally, the model generated rules relating to variability time , e.g., "If working days occur, then air pollution peaks in the morning and evening" and "If weekends occur, then pollution levels are generally lower." This indicates that the model can capture causality related not only to "static" relationships, but also dynamic temporal dependencies. This also means that the model can generate rules that express the multidimensionality of the underlying phenomenon (urban air quality).

In the case of seasonality, rules appeared such as: "If colder seasons occur, then air pollution levels increase due to heating demand" and "If warmer seasons occur, then ground-level ozone formation increases." These are examples of correct representation of secondary processes and meteorological relationships described in the text.

However, not all rules were correct. For example: " If and urban area exists , then human activities are influenced by it " is a simplification and partial inversion of dependencies. Similarly, general rules such as: " If air pollution exists , then it affects health ", although true, were not directly derived from the analyzed text, which indicates the phenomenon of "hallucination of an LLM".

Stage IV - Results of the. evaluation of the usefulness of language models using Bue metrics for the evaluation of association rules. In the fourth stage, a quantitative assessment was performed using the BLEU metric (see Table 1).

Table 1. Quantitative assessment was performed using the BLEU metric

Run	BLEU	Precision _1	Precision _2	Precision _3	Precision _4	Brevity_ Penalty	Length_Ratio	Translation_ Length	Reference_ Length
1_lama3.2_3b	0.07	0.24	0.08	0.05	0.03	0.90	0.91	427	471
2_lama3.2_3b	0.07	0.24	0.08	0.05	0.03	0.90	0.91	427	471
3_lama3.2_3b	0.00	0.13	0.02	0.01	0.00	0.85	0.86	318	371
4_lama3.2_3b	0.05	0.18	0.09	0.05	0.01	0.92	0.92	316	342
5_lama3.2_3b	0.04	0.21	0.08	0.04	0.01	0.83	0.84	363	431
1_mistral_latest	0.03	0.16	0.04	0.02	0.01	0.93	0.93	458	430
2_mistral_latest	0.03	0.17	0.04	0.03	0.01	0.89	0.90	399	444
3_mistral_latest	0.02	0.15	0.04	0.02	0.00	1.00	1.05	513	490
4_mistral_latest	0.02	0.15	0.04	0.02	0.00	1.00	1.05	513	490
5_mistral_latest	0.02	0.17	0.03	0.01	0.00	1.00	1.04	508	490
1_envgpt-q8_0_latest	0.00	0.17	0.01	0.00	0.00	0.82	0.83	406	487
2_envgpt-q8_0_latest	0.07	0.22	0.11	0.07	0.04	0.83	0.84	413	490
3_envgpt-q8_0_latest	0.07	0.21	0.11	0.07	0.03	0.84	0.85	418	490
4_envgpt-q8_0_latest	0.00	0.15	0.01	0.00	0.00	0.85	0.86	421	490
5_envgpt-q8_0_latest	0.00	0.15	0.01	0.00	0.00	0.85	0.86	421	490

The results indicated a low fit to the reference, which resulted from differences in sentence structure, element order, and rule detail, among other factors. For example, a semantically correct rule, but worded differently from the reference, received a low BLEU score.

Stage V - Results of the evaluation of the usefulness of language models using Rouge metrics for the evaluation of association rules. The fifth step involved analyzing ROUGE metrics. For the llama3.2_3b model, ROUGE-1 values were around 0.10, indicating partial content coverage. At the same time, ROUGE-2 values of 0.02–0.04 indicate poor matching of more complex linguistic structures, such as phrases like " vehicle emissions → nitrogen oxides ". Mistral_latest model achieved ROUGE-1 ≈ 0.06–0.07 scores, suggesting lower agreement with the source text. However, its responses were more uniform, e.g., it consistently followed the pattern " If X occurs , then Y increases, " which improved readability but not always accuracy.

Table 2. Quantitative assessment was performed using the Rouge metric

Run	ROUGE_1	ROUGE_2	ROUGE_L	ROUGE_LSum
1_lama3.2_3b	0.10	0.03	0.10	0.10
2_lama3.2_3b	0.10	0.04	0.10	0.10
3_lama3.2_3b	0.05	0.02	0.05	0.05
4_lama3.2_3b	0.08	0.02	0.08	0.08
5_lama3.2_3b	0.07	0.02	0.07	0.07
1_mistral_7b	0.07	0.02	0.06	0.06

2_mistral_7b	0.06	0.02	0.06	0.06
3_mistral_7b	0.07	0.01	0.06	0.06
4_mistral_7b	0.07	0.01	0.06	0.06
5_mistral_7b	0.07	0.02	0.06	0.06
1_envgpt-q8_0_latest	0.03	0.01	0.03	0.03
2_envgpt-q8_0_latest	0.13	0.05	0.12	0.12
3_envgpt-q8_0_latest	0.14	0.06	0.13	0.13
4_envgpt-q8_0_latest	0.04	0.01	0.04	0.04
5_envgpt-q8_0_latest	0.04	0.01	0.04	0.04

The envgpt-q8_0_latest model achieved the highest results in some runs, e.g., ROUGE-1 = 0.14 and ROUGE-2 = 0.06. In these cases generated more detailed rules, e.g., "If photochemical reactions occur under high solar radiation, then ground-level ozone concentration increases." However, in other runs, it generated very general rules, which lowered the results.

Stage VI - Prompt for a judge-type model to assess the suitability of language models. In the sixth stage, a judgement model was used for qualitative evaluation. The evaluation prompt required evaluation across eight criteria, including task relevance, completeness, and technical quality. This allowed for the capture of issues not identified by quantitative metrics.

Stage VII: Evaluation of the usefulness of association rule evaluation models on the prompt in stage VI. In the seventh stage, the cases of evaluation of language models using the judge model were analyzed. For example, for the case of the llama3.2_3b model, the answer contained only one rule: " If mobility , energy consumption and resources utilization occur , then atmospheric conditions change ". It received a rating of " overall_score : 4" and " verdict : fail ".

Model name	Context alignment	Text similarity	Completeness	Clarity	Sentence length style	Technical quality	Consistency	Safety	Overall score
Mistral 7B	7	5	5	8	8	7	8	10	6
llama 3.2_3b	6	4	5	7	7	5	6	10	5
envgpt-	7	5	5	8	8	7	8	10	6

q8_0_lat est									
-----------------	--	--	--	--	--	--	--	--	--

Detailed analysis showed that the model received a very low score for completeness (2) because it omitted most of the relationships present in the text, such as seasonal dependencies and measurement methods. The score for agreement with the text was also low (3) because the general concept of "atmospheric conditions" instead of specific pollutants.

At the same time, the model received higher scores for readability (5) and sentence structure (6), indicating that the problem lies not in language generation but in correct knowledge extraction. The technical score (4) indicates partial compliance with the format requirements, but lacks full prompt compliance. A comparison was also made between the models. Llama3.2_3b generated relatively correct but incomplete rules. Mistral_latest was more consistent but less accurate. EnvGPT showed the highest quality potential, but also the greatest result instability.

4 Discussion

The results align with existing literature on RAG systems, confirming that the quality of knowledge representation substantially influences model performance. The use of association rules contributes to improved interpretability, positioning the proposed approach within the broader explainable AI paradigm.

The application of an LMM as an LLM-as-a-Judge reflects current trends in automatic evaluation and offers a scalable alternative to human expert assessment. Moreover, the framework naturally supports future integration with knowledge distillation techniques, where the selected model may act as a domain-specific teacher.

The primary limitations of this study include dependence on the quality of the input corpus and the inherently subjective nature of LMM-based evaluation. Future research will focus on incorporating multiple evaluators (LMM ensembles), analyzing evaluation stability, and integrating explainability mechanisms at the response level.

The analysis of evaluation scores revealed a consistent advantage of one model LLM-2, particularly in terms of factual correctness and analytical usefulness. LLM-1 demonstrated strong linguistic coherence, whereas LLM-3 exhibited the highest variability in response quality. This general pattern was observed consistently across the evaluated query set and formed the basis for the more detailed analysis presented below. All models were evaluated under identical Retrieval-Augmented Generation conditions, relying on the same association-rule-based knowledge representation and vector retrieval mechanism. Consequently, performance differences can be primarily connected to model-specific generation and reasoning capabilities rather than to variations in retrieval, input data or judgement by another reasoning model.

For the initial quantitative analysis BLEU and ROUGE scores were computed by comparing generated outputs against a predefined reference document representing the association rules in the desired format from the expertise. As anticipated at an early

stage of the analysis, both metrics generally reached low absolute scores for all models. While minor variations between the evaluated models were observable, these differences were neither stable nor sufficient enough to support a reliable ranking.

In contrast, the LLM-as-a-Judge evaluation produced a clear and consistent differentiation between the evaluated models. Across the majority of evaluated queries, LLM-2 achieved the highest overall scores, emerging as the most consistently well performing model. Judge-based evaluation highlights LLM-2 ability to preserve the causal and conditional structure within the association rules. It also manages to translate this knowledge into coherent and precise responses. Its outputs were frequently higher rated as complete. In addition to that it accurately covers the necessary facts without introducing unsupported concepts or excessive abstraction. From assessing the Model outputs manually it seemed to have stayed true to the format over all test runs.

The remaining models showed different, but also less satisfactory, results. The LLM-1 created significantly differing association rules and diverges from the reference content. On the other hand the LLM-3 demonstrated strong linguistic features and appears to structure the sentences well but more frequently omitted relevant rule components or made new rules up, resulting in lower scores for completeness and technical precision. Also the model started to add contextual information of the received documents which also went against the rules set in the generation prompt.

Through the qualitative analysis of the LLM-as-a-Judge and the justifications generated it further reinforced the findings. The judge consistently awards higher scores to models that fully meet the expectations regarding the properties of the association rules and can represent them logically and coherently. Lower scores usually correlate with a format violation or the invention / derivation of association rules based on the provided context. Even differences in the rule or sentence structure, or the order of the content, have been sufficient grounds for the judge to award lower scores. These findings particularly stand out in the judge's concluding summary of all properties or the overall score given.

Based on the conducted experiment, a moderately critical assessment can be formulated regarding the suitability of the analyzed language models for air quality data analysis tasks. The results of automatic evaluation metrics, particularly BLEU (0–0.07) and relatively low ROUGE scores (ROUGE-1 predominantly in the range of 0.05–0.14), indicate limited alignment between generated outputs and reference descriptions. This suggests that the models struggle to precisely reproduce the expected analytical content, which constitutes a significant limitation in scientific applications requiring high semantic fidelity.

At the same time, the expert-based evaluation (judge model) provides a more nuanced perspective. The models Mistral 7B and envgpt-q8_0_latest achieve comparable and relatively higher overall scores (6/10), outperforming llama3.2_3b (5/10). Notably, high ratings in categories such as clarity, sentence length style, and consistency (7–8 points) indicate that the generated texts exhibit good linguistic fluency and narrative coherence. However, moderate scores in text similarity and completeness (around 5 points) confirm the limited ability of the models to fully and accurately reproduce source information. The consistently high score in the safety category (10/10 for all

models) highlights their robustness and the absence of harmful or undesirable content generation.

From a comparative perspective, `envgpt-q8_0_latest` demonstrates the highest variability in ROUGE scores, achieving both the best results in the experiment (ROUGE-1 = 0.14) and some of the lowest, which may indicate generative instability. Mistral 7B shows more stable, albeit moderate, performance, while `llama3.2_3b` yields the weakest results across both automatic metrics and qualitative evaluation.

5 Conclusions

The proposed framework provides a formal and reproducible approach for selecting language models in RAG systems applied to environmental protection data. By combining association-rule-based knowledge representation, vector retrieval, and LLM-as-a-Judge evaluation, the method enables informed model selection for high-stakes, domain-specific applications.

The following conclusions summarize the comparative performance of the evaluated language models in the context of air quality data analysis. They highlight key strengths, limitations, and practical implications for their application in scientific and analytical workflows:

1. Low BLEU and ROUGE values indicate that all models have limited capability to accurately reproduce reference analytical content.
2. The models demonstrate relatively strong linguistic quality (clarity and style), making them useful for generating narrative descriptions.
3. Mistral 7B and `envgpt-q8_0_latest` achieve comparable overall performance and outperform `llama3.2_3b`.
4. `envgpt-q8_0_latest` exhibits the highest variability in results, which may reduce its reliability in production scenarios.
5. Mistral 7B provides the most stable trade-off between quality and consistency.
6. `llama3.2_3b` shows the lowest suitability in the analyzed scenario, particularly in terms of technical quality and consistency.
7. High safety scores suggest that all models can be safely deployed in environmental data processing contexts.
8. The results indicate that the evaluated models are better suited as supportive tools for data interpretation rather than as precise analytical systems requiring high fidelity to reference data.

References

1. Mousavian Anaraki, S. A., Croce, D., Basili, R.: Large language models for sustainability reporting: A systematic review and research agenda. *Sustainable Futures*, 10, (2025)
2. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.-T., Rocktäschel, T., Riedel, S. and Kiela, D.: Retrieval-augmented generation

- for knowledge-intensive NLP tasks, *Advances in Neural Information Processing Systems*, 33, (2020)
3. Izacard, G. and Grave, E.: Leveraging passage retrieval with generative models for open domain question answering', *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, (2021)
 4. Croxford, E., Gao, Y., First, E., Pellegrino, N., Schnier, M., Caskey, J., Oguss, M., Wills, G., Chen, G., Dligach, D., Churpek, M. M., Mayampurath, A., Liao, F., Goswami, C., Wong, K. K., Patterson, B. W., Afshar, M.: Evaluating clinical AI summaries with large language models as judges. *npj Digital Medicine*, 8, (2025)
 5. Baysan, M. S., Uysal, S., İşlek, İ., Çığ Karaman, Ç., Güngör, T.: LLM-as-a-Judge: automated evaluation of search query parsing using large language models. *Frontiers in Big Data*, 8, (2025)
 6. Rudin, C.: Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1(5), (2019)
 7. Es, S., James, J., Anke, L.E., Schockaert, S.: Raga's: Automated evaluation of retrieval augmented generation. *Proceedings of the 18th conference of the European chapter of the association for computational linguistics: system demonstrations*, (2023)
 8. Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Wang, H.: Retrieval-augmented generation for large language models: A survey. *arXiv* (2023)
 9. Truong, T. H., Le, M. T., Nguyen, H. T.: Benchmarking evaluation methods for retrieval-augmented generation systems. *Proceedings of the AAAI Conference on Artificial Intelligence*, (2024)
 10. Bhuyan, B. P., Ramdane-Cherif, A., Tomar, R., Singh, T. P.: Neuro-symbolic artificial intelligence: a survey. *Neural Computing and Applications*, 36(21), (2024)
 11. Garcez, A. d'Avila, Lamb, L. C., Gabbay, D. M.: *Neural-symbolic cognitive reasoning*. Springer, (2019)
 12. Besold, T. R., d'Avila Garcez, A., Bader, S., Bowman, H., Domingos, P., Hitzler, P., Lamb, L. C.: *Neural-symbolic learning and reasoning: A survey and interpretation*. *arXiv*, (2017)
 13. Li, H., Su, Y., Cai, D., Wang, Y., Liu, L.: A survey on retrieval-augmented text generation, *arXiv*, (2022)
 14. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Riedel, S.: Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems*, (2020)
 15. Izacard, G., Grave, E.: Leveraging passage retrieval with generative models for open domain question answering. *arXiv preprint arXiv:2007*, (2021)
 16. Kadavath, S., Conerly, T., Askell, A., Henighan, T., Drain, D., Perez, E., Kaplan, J.: Language models (mostly) know what they know, *arXiv:2207*, (2022)
 17. Jiang, Z., Zhou, F., Neubig, G., Chen, Y.: How can we know what language models know? *Transactions of the Association for Computational Linguistics*, 9, (2021)
 18. Lin, S., Hilton, J., Evans, O.: TruthfulQA: Measuring how models mimic human falsehoods. *arXiv preprint arXiv:2109*, (2022)